



Xiaohua Semiconductor MCU Development Environment Usage

This product supports the following chip series:

All series	All models
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- This manual takes the HC32L110 series as an example. If you have any questions, please feel free to contact us. Contact information is on the back cover.

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1. Overview

This application note will explain how to use the sample projects provided by Xiaohua Semiconductor for software development in the Keil MDK and IAR Workbench environments.

2. Software Development

2.1 Tool Options

Xiaohua Semiconductor MCUs are all designed based on the Cortex-M0+/M4 core, and can be used for software development in a variety of third-party development environments. The official driver library and sample projects are mainly based on the following two development tools:

- IAR Embedded Workbench for ARM
- Keil ARM RealView® Microcontroller Development System

The evaluation version and detailed usage information can be downloaded from its official website.

- <https://www.iar.com/iar-embedded-workbench/#!?architecture=Arm>
- <http://www2.keil.com/mdk5>

2.2 Obtaining Driver Libraries and Sample Code

Xiaohua Semiconductor provides users with driver libraries and sample codes corresponding to each MCU model to support users to quickly get started and shorten product development time.

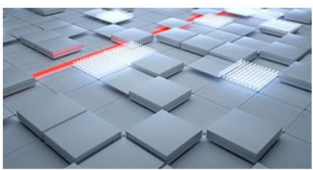
Users can download the required code from the official website of Xiaohua Semiconductor. Take HC32L110C6PA-TSSOP20 as an example:

1. Enter the official website of Xiaohua Semiconductor: <http://www.xhsc.com.cn>
2. Click "Ultra-low power MCU" in "Product Series"

产品系列



超低功耗MCU



通用类MCU



电机类MCU



车规MCU

3. Click to enter "HC32L110C6PA-TSSOP20"

产品型号	主频 (MHz)	内核	FLASH (KB)	RAM (KB)	EE (Byte)	GPIO	工作电压 (V)	DMA	低功耗 定时器	基本 定时器	通用 定时器	高级 定时器
HC32L110C6PA-TSSOP20	32	M0+	32	4	0	16+1	1.8 - 5.5	-	1	0	6	0
HC32L110C6PA-TSSOP20TR	32	M0+	32	4	0	16+1	1.8 - 5.5	-	1	0	6	0
HC32L110C6UA-SFN20TR	32	M0+	32	4	0	16+1	1.8 - 5.5	-	1	0	6	0

4. Select the "Development Tools" column to download the driver library, samples, and IDE support package

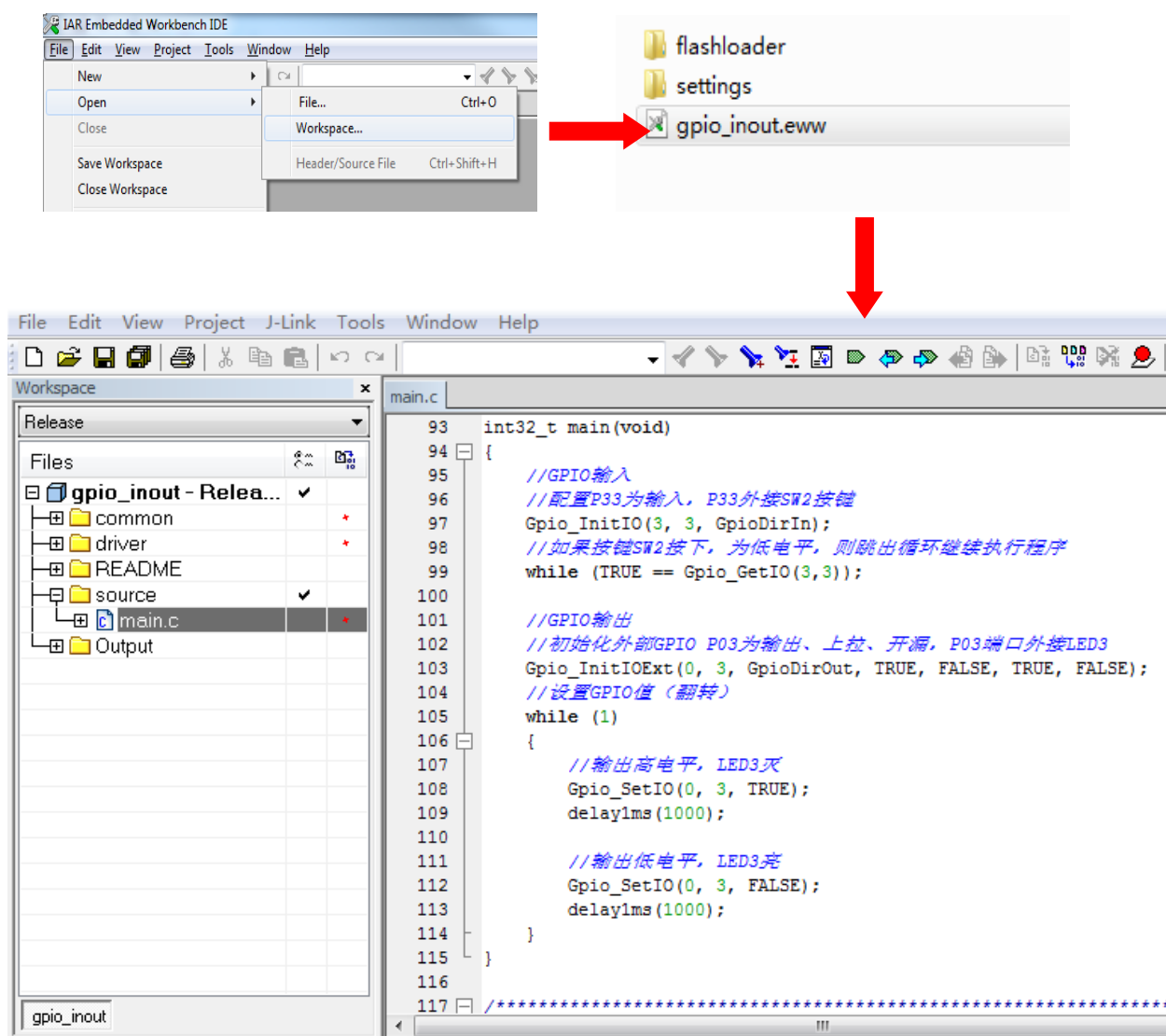
产品特点	技术文档	开发工具	应用方案
华大开发板			
STK-HC32L110_F003_005-TSSOP20 PDF BOM.zip			
zip 2020-08-12			
STK-HC32L110_F003_005-TSSOP20设计文件.zip			
zip 2020-08-12			
HC32L110_F003_005原理图库文件.zip			
zip 2020-08-12			
驱动库及样例			
HC32L110_DDL_Rev1.1.4.zip			
zip 2020-07-31			
HC32L110_TEMPLATE_Rev1.0.2.zip			
zip 2020-07-31			
IDE支持包			
HC32L110_IDE_Rev1.0.3.zip			
zip 2020-07-31			

2.3 Using IAR Workbench to Open Sample Projects

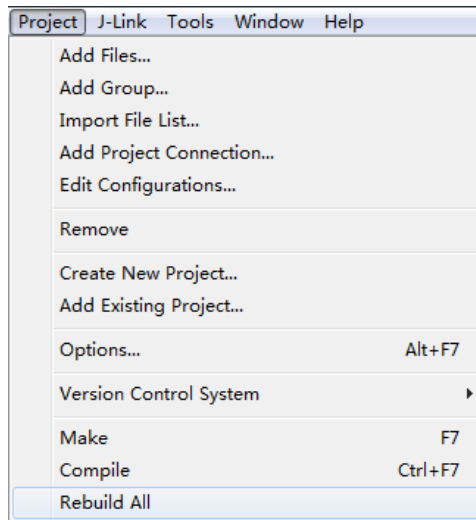
The following steps will describe how to open, compile, run and debug the sample code in the IAR Workbench environment.

1. Unzip the downloaded DDL
2. Open IAR Embedded Workbench IDE V7.70 (or higher)
3. Click File->Open->Workspace and select gpio_inout.eww from the directory

<Storage directory>: \HC2L110_DDL_Rev1.1.4\example\gpio\gpio_inout\EWARM



4. Click Project->Rebuild All to compile the project

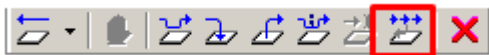


5. Configure the driver, power supply, etc. according to the user manual or quick manual of the corresponding development board, and then power it on via the USB cable to ensure that there is no abnormality in the circuit board.

6. Click in the function  or the shortcut key Ctrl+D to download the program



7. Click  to start the program running



8. Click  to stop the program running

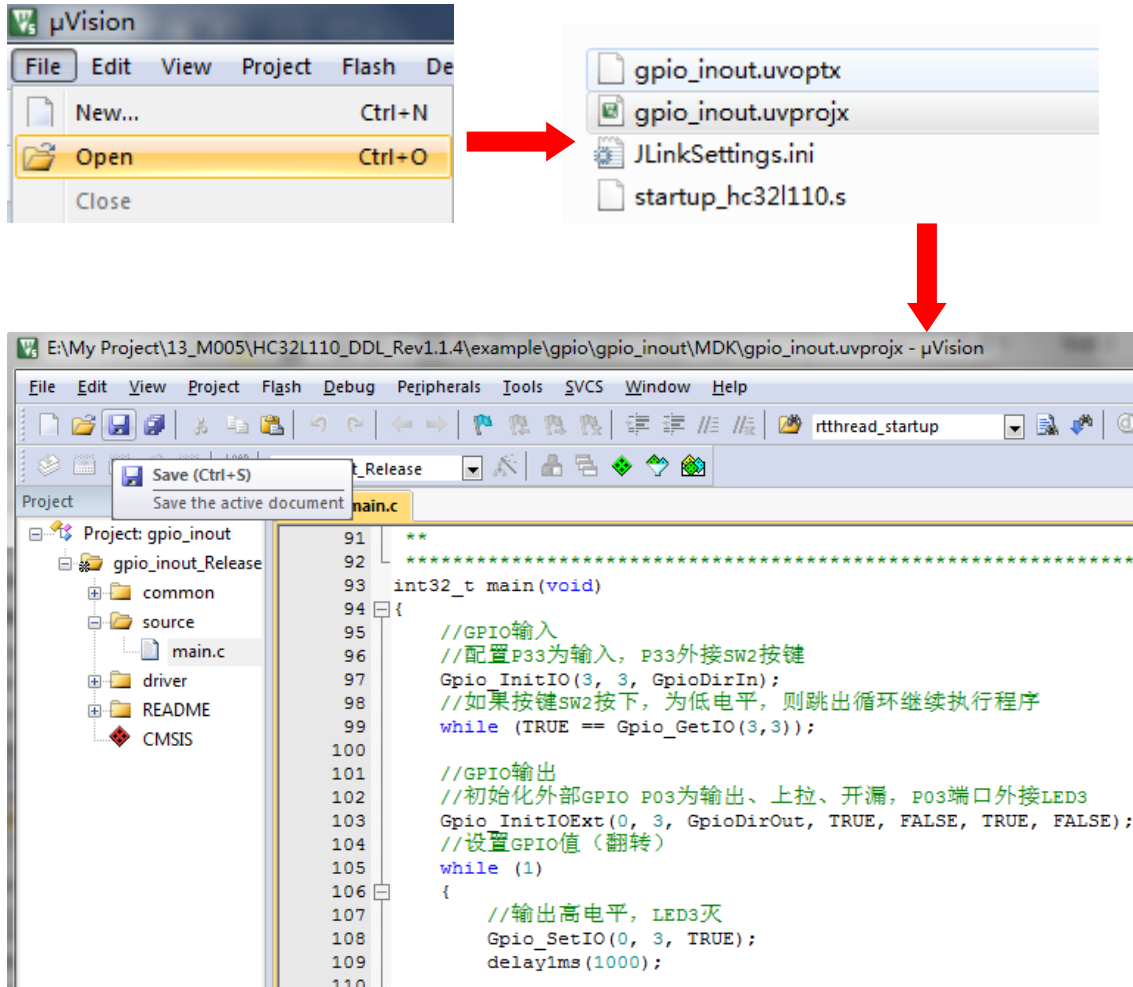


For more detailed information about IAR Workbench, please click the Help column.

2.4 Using Keil uVision IDE to Open Sample Projects

The following steps will introduce how to open, compile, run and debug sample code in Keil uVision IDE environment.

1. Unzip the downloaded DDL
2. Open Keil uVision IDE v5.11a (or higher)



3. Click  to compile project



4. Configure the driver, power supply, etc. according to the user manual or quick manual of the corresponding development board, and power it on through the USB cable to ensure that there is no abnormality in the circuit board.

5. 点击  下载程序



5. Click  to download program



7. Click  to stop the program



2.5 Using Templates to Build Your Own Projects

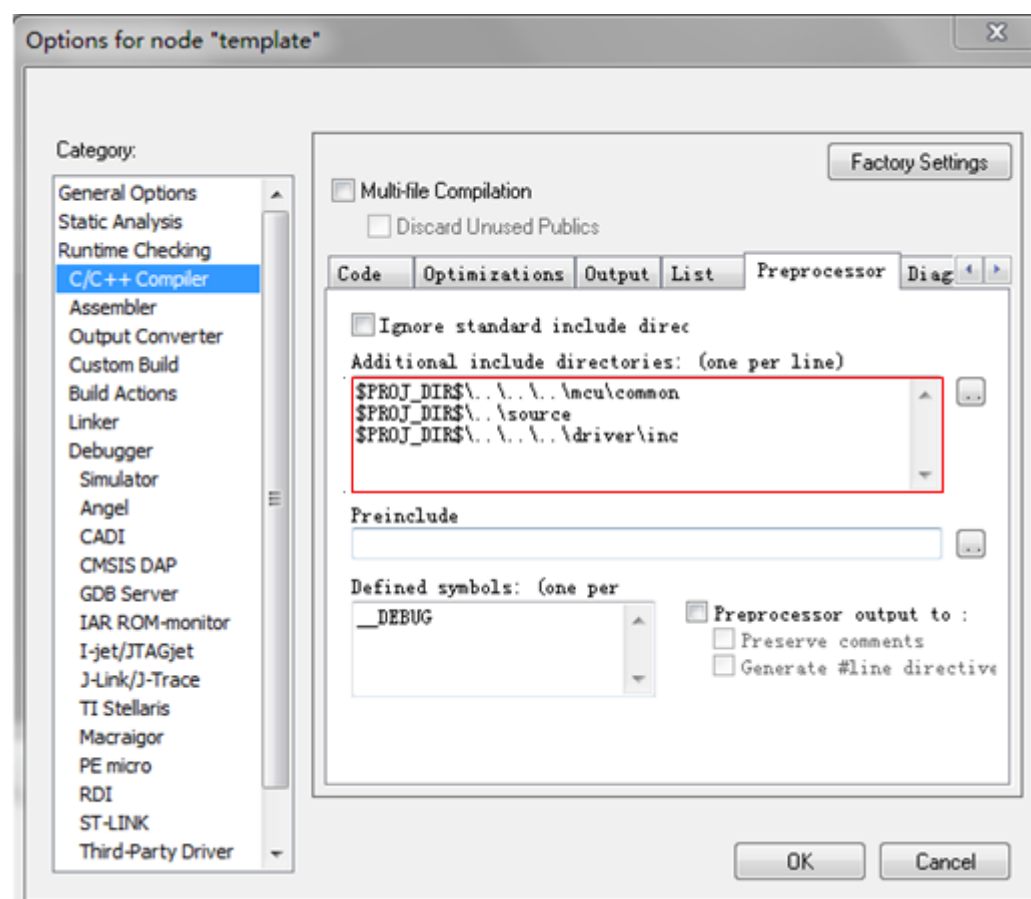
For new projects created by users, it is recommended to modify the template in the driver library provided by Xiaohua Semiconductor and add the user's own files.

Template path:

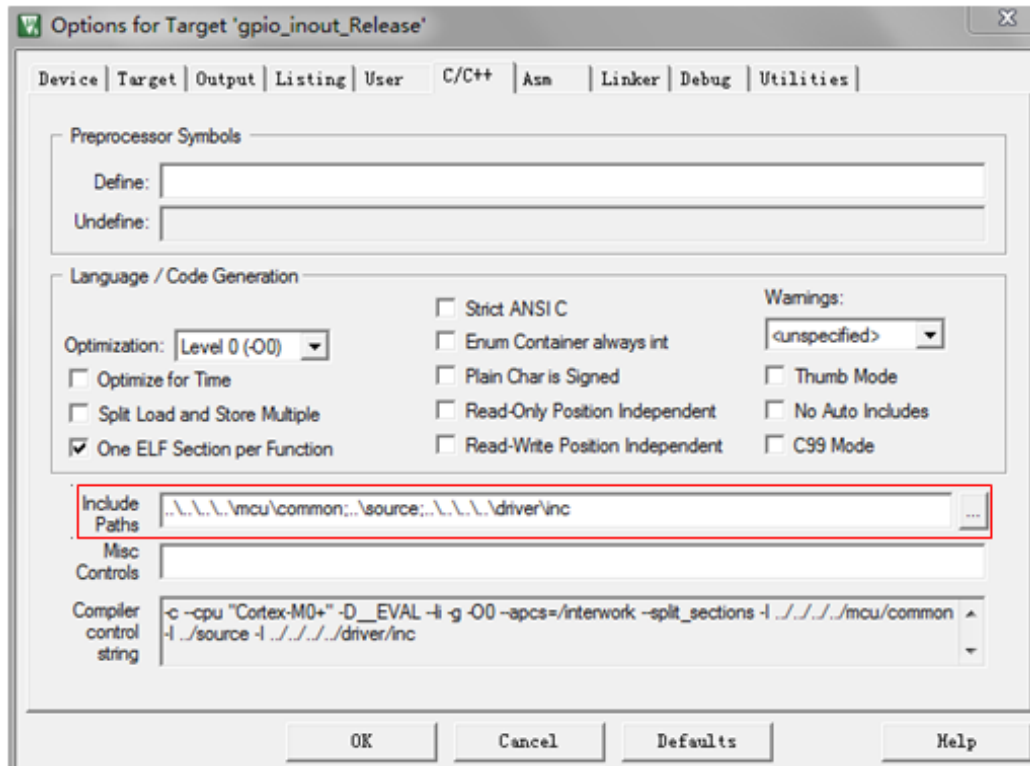
<Storage directory>: HC32L110_DDL_Rev1.1.4\example\template

It should be noted that the header file path of the project is set based on the file structure of the entire DDL. If the user modifies the structure of the template and other folders, the header file inclusion path must also be modified accordingly, otherwise a compilation error will occur.

➤ IAR Workbench project



➤ Keil uVision Project



If the customer really wants to create a new project, he can refer to the user manual of the corresponding development environment. And download the IDE support package for the corresponding MCU.

2.6 FAQs

1. Cannot find the chip

Solution:

- Download the "IDE support package" according to 2.2, unzip it and copy it to the corresponding folder of the development tool.
- IAR Workbench

Copy the folders in IAR_IDE->config to the folders corresponding to IAR Workbench

<Save Directory>: HC32L110_IDE_Rev1.0.3\HC32L110_IDE_Rev1.0.3\IAR_IDE\config

<Installation Directory>: IAR Systems\Embedded Workbench 7.5\arm\config

- Keil uVision IDE

Click the .pack file in the following file to install

<Save Directory>: HC32L110_IDE_Rev1.0.3\HC32L110_IDE_Rev1.0.3\MDK_IDE

Note that please confirm the installation path during installation

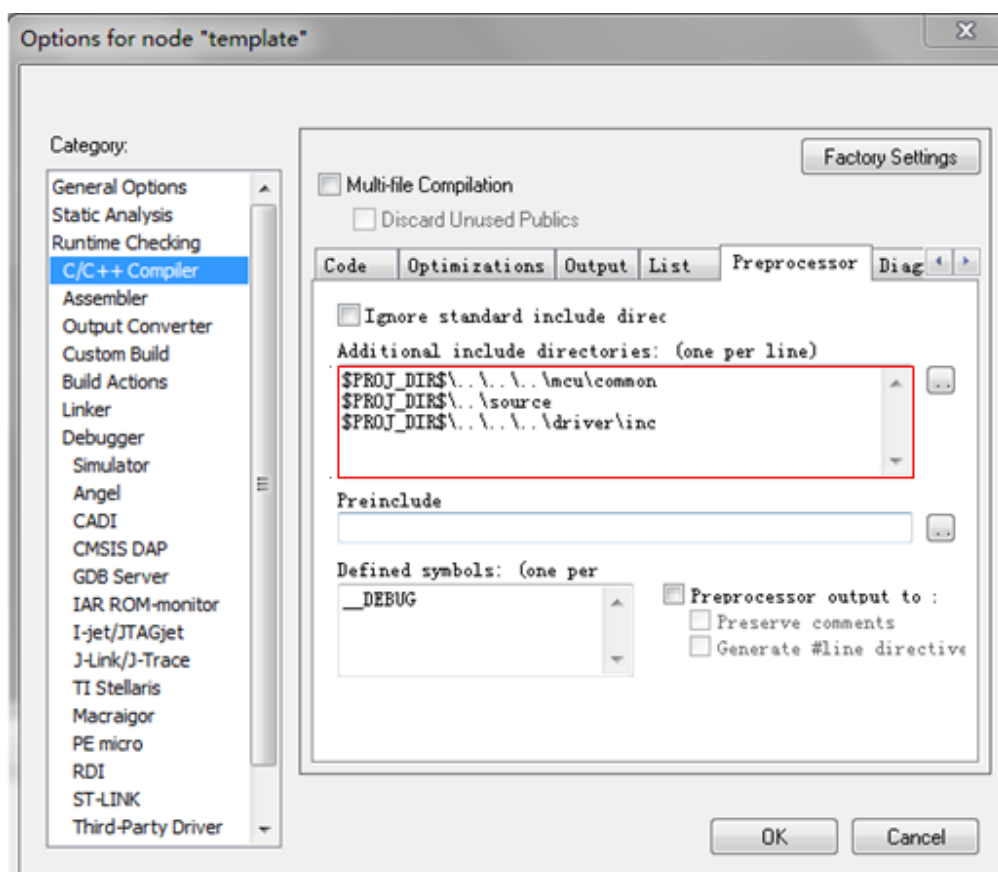
<Keil Installation Path>: ARM\PACK

2. Cannot find .h file

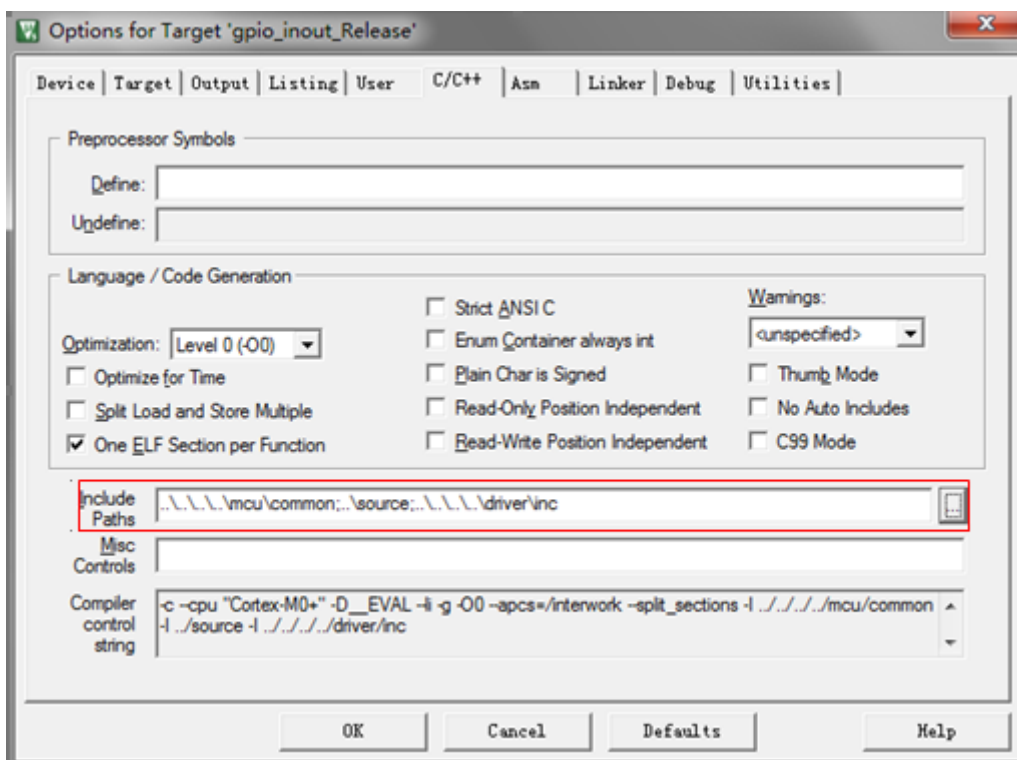
.h file missing or include path

Solution:

- Confirm that the .h file exists and ensure that the include path is correct
- IAR Workbench



➤ Keil uVision IDE



3. Version Information & Contact Us

Date	Version	Change History
2020/11/20	Rev1.0	First version released.
2022/7/15	Rev1.1	Company logo updated.



If you have any comments or suggestions during the purchase and use process, please feel free to contact us.

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